

Question				Marking details			Marks available														
							AO1	AO2	AO3	Total	Maths	Prac									
1	(a)	(i)		{tRNA / transfer RNA / transfer} NOT transport RNA and {rRNA / ribosomal RNA / ribosomal} (1)			1			1											
		(ii)		<table><tr><td></td><td>Bacterial cell</td><td>Animal cell</td></tr><tr><td>Location(s) of ribosome involved in translation</td><td>Cytoplasm</td><td>rough endoplasmic reticulum (and cytoplasm)</td></tr><tr><td>Size of ribosomes</td><td>70S / small(er)</td><td>80S/ large(r)</td></tr></table> 2 marks for 4 correct boxes 1 mark for 2 or 3 correct boxes 0 marks for 1 correct boxes				Bacterial cell	Animal cell	Location(s) of ribosome involved in translation	Cytoplasm	rough endoplasmic reticulum (and cytoplasm)	Size of ribosomes	70S / small(er)	80S/ large(r)	2			2		
	Bacterial cell	Animal cell																			
Location(s) of ribosome involved in translation	Cytoplasm	rough endoplasmic reticulum (and cytoplasm)																			
Size of ribosomes	70S / small(er)	80S/ large(r)																			
	(b)			{Animal cell / it} {is larger / has a larger surface) (than bacterial cell) (1) {Animal cell / it} contains { <u>membrane</u> bound / named} organelles, (bacterial cell doesn't) / ORA (1)				2		2											
	(c)	(i)		A Phospholipids, + B (intrinsic / channel / carrier / transmembrane) Proteins (1) A physical barrier / {diffusion / transport} of {small / non-polar / owtte / or example} molecules (1) B facilitated diffusion / active transport / transport of {large / polar molecules/ions / owtte / or example} (1) If label incorrectly can still access function marks if they match the structure given For mark point 2 and 3 allow named examples of substances moved			3			3											

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		(ii)	I	no need for {recognition / receptor / or description of}			1	1		
			II	Any one (x1) from <u>More</u> {facilitated diffusion / active transport / or description of} (1) transport of ATP (1) attachment of enzymes (1) Accept reference to presence of {proton pumps/ ATP synth(et)ase}			1	1		
				Question 1 total	6	2	2	10		